Pre-analysis plan: #manydaughters
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1 Interpretation of the hypotheses

In this section, we clarify how we interpret the #manydaughters hypotheses. We will use H1 as an example but the reasoning applies analogously to H2, H3, and H4. H1 says:

> *"Having daughters leads to stronger support for gender equality (as measured by attitudes).*
> *That is, denoting X as a measure of support for gender equality,*
> *(X for respondents with daughters) > (X for respondents without daughters)."*

Taking this hypothesis literally, it means that parents' support for gender equality is stronger after the birth of their daughter than before. Strictly speaking, the hypothesis does NOT set the change in gender equality support for parents of daughters in relation to the change in gender equality support for the parents of sons. I.e., the hypothesis would still be confirmed if having daughters leads to the same increase in support for gender equality as having sons. However, based on the previous literature (e.g., Cronqvist and Yu 2017; Borrell-Porta, Costa-Font, and Philipp 2019; Yang et al. 2025), we believe that the hypothesis is meant to be interpreted as regarding the effect of having daughters compared to having sons.

Similarly, taking "(X for respondents with daughters) > (X for respondents without daughters)" literally would mean that we compare gender equality support between respondents that have daughters and all other respondents (i.e., also respondents that do not have any children at all). Again, we believe that this is not the hypothesis that we are meant to test, given the additional remark that "RTs shall consider these four research hypotheses that they are testing as causal hypotheses when developing their analysis". A simple comparison of gender equality support between respondents with and without daughters would not let us test the causal effect of having daughters as respondents with daughters are likely systematically different from respondents without daughters in several other aspects relevant for gender equality support, such as family-oriented values that drive the decision to have children in the first place.

Moreover, it remains to be clarified whether the hypothesis relates to the effect of having any daughters or the effect of having an additional daughter, both of which have been investigated in previous studies (e.g., Yang et al. 2025). In this regard, we believe that the remark "(X for respondents with daughters) > (X for respondents without daughters)" is supposed to tell us that we should test the effect of having any daughters.

In sum, we interpret the hypothesis as follows: "A given set of parents has a stronger support for gender equality in a universe where chance gives them at least one daughter compared with a universe in which chance gives them only sons, ceteris paribus."

2 Regression design

In this section, we present our regression design. In line with our interpretation of the hypotheses, we will use instrumental variables (IV) regressions, where the dependent variables are measures of support for gender equality/political preferences outlined below and the main independent variable is a dummy that is one if a respondent has at least one daughter (zero otherwise), instrumented by a dummy that is one if the firstborn child is a daughter (zero otherwise). Intuitively, we thereby estimate the effects of having daughters as the differences in gender support/political preferences between parents whose firstborn is a daughter and parents whose firstborn is a son scaled by the difference in the rate of having any daughter between parents whose firstborn is a daughter and parents whose firstborn is a son. The scaling accounts for the fact that not every parent whose firstborn is a son does not have any daughter. E.g., let us assume the true effect of having daughters is that gender equality support rises from 0 to 2. Moreover, we assume that 40% of parents whose firstborn is a son still have a daughter. Then, the difference in gender equality support between parents whose firstborn is a daughter and parents whose firstborn is a son is $2 - (0.4 * 2 + 0.6 * 0) = 1.2$. Scaling this difference by the difference in the rate of having any daughter between parents whose firstborn is a daughter and parents whose firstborn is a son $(100\% - 40\% = 60\%)$ yields the true effect of $1.2/60\% = 2$.

A similar IV regression design is outlined by Washington (2008) who suggests instrumenting the final gender mix with the gender of the firstborn, although the author does not implement the design due to a lack of data on the gender of the firstborn. The reason for instrumenting the any-daughter dummy with the firstborn-is-daughter dummy is that the gender of the firstborn is exogenous while having any daughter is endogenous to fertility stopping rules which may, e.g., be driven by a preference for having sons that is plausibly related with gender equality support and political preferences (e.g., Washington 2008; Oswald and Powdthavee 2010; Conley and Rauscher 2013; Lee and Conley 2016; Cronqvist and Yu 2017; Borrell-Porta, Costa-Font, and Philipp 2019; Clayton, de Kadt, and Dumas 2023).

For example, let us imagine there are 100 conservatives and 100 liberals, each of whom can have a maximum of two children. For the sake of the example, assume that political

preferences determine fertility stopping rules. Conservatives follow the fertility stopping rule of procreating until they have a son (or reach the maximum of two daughters). Liberals always produce 2 children. Political preferences are completely stable, i.e., children and their gender have no effect on political preferences. The probabilities for sons and daughters are 50/50. In such a world, 50 conservatives would first have a son and then stop. 25 (25) conservatives would first have a daughter and then a son (daughter). 25 (25; 50) liberals would have two sons (two daughters; a son and a daughter). Among respondents with at least one daughter, there are 75 liberals and 50 conservatives, i.e., 60% are liberal. Among respondents without a daughter, there are 25 liberals and 50 conservatives, i.e., 33% are liberal. Therefore, although having daughters has no effect on political preferences, regressing political preferences on an any-daughter dummy without IV would falsely indicate that having at least one daughter increases liberal preferences by 27pp. Controlling for the number of children would not help: Among respondents with two children and at least one daughter, there are 75 liberals and 50 conservatives, i.e., 60% are liberal. Among respondents with two children and without a daughter, there are 25 liberals and 0 conservatives, i.e., 100% are liberal. Regressing political preferences on an any-daughter dummy without IV would falsely indicate that having at least one daughter decreases liberal preferences by 40pp. However, comparing only the gender of first children, the rate of liberal preferences is 50% among respondents for which the firstborn child is a daughter and also 50% among respondents for which the firstborn child is a son, yielding the correct effect of zero in our proposed IV regression.

While in the scenario above, including respondent fixed effects in a regression of political preferences on the any-daughter dummy would also correctly estimate an effect of zero, respondent fixed effects do not protect against time-varying endogeneity within a respondent. If, e.g., becoming more liberal drives a respondent to abandon his stop-at-first-son rule, having daughters is associated with becoming more liberal when using a fixed effects regression to study the within-respondent variation in political preferences in relation to having daughters, even though there is no effect of having daughters on political preferences. Again, our proposed IV regression is not affected by this reverse causality because our instrument, the gender of the first child, is exogenous.

Another consideration in the regression design is whether to weight observations to make the sample representative of the population in terms of characteristics such as age, gender, region, etc. This is necessary if, e.g., certain demographic groups are underrepresented in the sample because they have an increased probability of not responding to the survey. Studying the effect of having daughters on political preferences in the European

Social Survey, Yang et al. (2025) use weights which are calibrated to make the European Social Survey sample representative of the European population. The SOEP contains analogous weights which are calibrated to make the SOEP sample representative of the German population. The problem with using these weights in our analyses is that the weights are calibrated to make the *whole* SOEP sample representative of the *whole* German population, while we would need weights that make the *sub*sample of SOEP parents representative of the *sub*population of German parents.

The best solution would be to create such "subsample weights" ourselves but unfortunately we lack population-level data on the characteristics of German parents. Therefore, we are left with the options to use the "full-sample weights" provided in the SOEP data or to use no weights at all. Using full-sample weights on the subsample of parents would only be appropriate if the associations between the characteristics (age, gender, region, etc.) and the probability of being in the SOEP sample were the same for parents and non-parents which we believe to be a strong assumption. If this assumption is violated, it is not clear whether the weighted subsample of parents is more or less representative of the subpopulation of parents than the unweighted subsample of parents.

The following stylized example illustrates a situation in which it is better to use an unweighted subsample: There are old and young individuals with and without children. Old individuals always respond while young individuals only respond when they have children. Population: 50 old with child, 50 old w/o child, 50 young with child, 50 young w/o child. Sample: 50 old with child, 50 old w/o child, 50 young with child, 0 young w/o child. Weights for individual respondents (Equation 3) are calibrated s.t. the sample is representative of the population in terms of 50% old and 50% young individuals (Equation 1) and s.t. the weights sum to one (Equation 2). Parent-status is not considered in the calibration of the weights.

$$\frac{100w_{Old}}{100w_{Old} + 50w_{Young}} = 0.5 \tag{1}$$

$$100w_{Old} + 50w_{Young} = 1 \tag{2}$$

$$w_{Old} = 0.005; w_{Young} = 0.01 \tag{3}$$

Share of old individuals among parents in population: $\frac{1}{2}$.
Share of old individuals among parents in weighted sample: $\frac{50 \cdot 0.005}{50 \cdot 0.005 + 50 \cdot 0.01} = \frac{1}{3}$.
Share of old individuals among parents in unweighted sample: $\frac{50}{50+50} = \frac{1}{2}$.

Of course, it depends on one's assumptions whether using no weights is better than using full-sample weights and if we could run sensitivity analyses, we would simply do both, as recommended by Solon, Haider, and Wooldridge (2015). Yet, given that we can only run one specification and that we have no prior beliefs about which assumptions are warranted, we apply Occam's razor and choose the simpler approach of using no weights.

3 Variables

3.1 H1/H4 measure for support for gender equality (in terms of attitudes)

Variable that we will use: pl/plh0308_v2: "On a scale of 1–7, how much do you agree with the following statement: It's best if the man and the woman do the same amount of paid work and take care of housework and family to the same degree." (literal translation, SOEP translation: "It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally.").
Recorded in 2018 with 25,570 valid responses.
We will use this variable because it captures the essence of supporting gender equality in the context of sharing the different duties relating to paid work, housework, and family life equally between men and women. In the literature, similar measures are used for attitudes towards supporting gender equality (see Table 1).

We will use the 1–7 values as a linear variable without transformation. In this regard, Andrew Gelman writes in his blog (https://statmodeling.stat.columbia.edu/2008/10/30/more_on_scaling/): "Question by Tom Knapp: ... I think Likert scales are inappropriate for linear regression analysis in general and require the use of ordinal regression analysis. Answer by Andrew Gelman: I'm not so bothered by treating a 1–5 or 1–10 scale linearly, on the assumption that the difference between 1 and 2 is approximately the same as the difference between 3 and 4, or whatever."

Variables that we considered but will not use: pl/plh0308_v1: "On a scale of 1–4, how much do you agree with the following statement: It's best if the man and the woman do the same amount of paid work and take care of housework and family to the same degree." (same as above).
Recorded in 2012 with 7,072 valid responses.

We will not use this variable because it is not recorded for the main SOEP sample but for the "Families in Germany" sample which is not a representative sample but instead selects single parents, families with at least three children, low-income families, and families with newborns from 2007–2010 (see https://www.diw.de/documents/publikationen/73/diw_01.c.805759.de/diw_ssp0905.pdf). The same goes for other pl/plh030x variables, such as pl/plh0304 ("Children suffer when the father is focused on his career.").

pl/plh0302_v2 (pl/plh0298_v2): "On a scale of 1–7, how much do you agree with the following statement: Children under the age of 3 (6) suffer when the mother works.".
Recorded in 2018 with 25,544 (25,527) valid responses.
We will not use these variables as they do not necessarily reflect attitudes regarding gender equality. Respondents that agree with these statements could still support gender equality, e.g., if they also agree that children suffer when the father works.

pl/plj0620 ("In a democracy, women should have the same rights as men") – pl/plj0625 ("At home, the husband should have the final say.").
Recorded from 2016–2018 with around 7,000 valid responses per question across all years.
We will not use these variables because they were not recorded for the main SOEP sample but only for the sample of asylum seekers (see, e.g., https://paneldata.org/soep-core/instruments/soep-core-2018-pb-m345-erst).

3.2 H2 measure for left-leaning political preferences

Variable that we will use: pl/plh0004: "In politics people often talk about left and right when it comes to characterizing different political attitudes. If you think about your own political views: Where would you place yours on a scale of 0 (left) to 10 (right)?"
Recorded in 2005, 2009, 2014, and 2019 with a total of 90,972 valid responses.
We will use this variable because self-placements between left and right fit the hypothesis well and are a common measure in the recent literature (see Table 2). As for H1/H4, we will use the 0–10 values as a linear variable without transformation.

Variables that we considered but will not use: Oswald and Powdthavee (2005), Oswald and Powdthavee (2010), and Clayton, de Kadt, and Dumas (2023) use support for left-wing vs. right-wing political parties to measure political preferences. With SOEP data from 1985–2002, Oswald and Powdthavee (2005) find confirmatory evidence for H2 using support for the traditional left-wing party SPD vs. the traditional right-wing party CDU/CSU as their binary dependent variable and the number of daughters that a

person has as their independent variable. The authors estimate that for every daughter the probability that a man votes for the left-wing SPD (as opposed to the right-wing CDU/CSU) increases by 2.5pp in a linear probability model with individual fixed effects. For women, the authors find no effect. Based on our understanding of the methodology used in Oswald and Powdthavee (2005), the point estimates rely on 638 switches from SPD to CDU/CSU, and 660 switches from CDU/CSU to SPD between two consecutive survey waves, with 47,202 respondents not switching party support between survey waves. We prefer to use self-placements between left and right on a 0–10 scale rather than party support as our measure for left-leaning political preferences because we believe that the scale is more granular and less sensitive to a changing party landscape. Regarding granularity, there is a spectrum between left and right within each party which can be represented on the 0–10 scale but which goes missing when studying binary party support, leading to a loss of variation in political preferences, which is reflected in the small numbers of switches in Oswald and Powdthavee (2005). Regarding a changing party landscape, while CDU/CSU and SPD together account for a large fraction of party preferences during the 1985–2002 sample that Oswald and Powdthavee (2005) study, their support has decreased over time, giving rise to a more splintered party landscape that is less easy to judge in terms of which party is more/less left vs. right (see https://www.bundestag.de/parlament/wahlen/ergebnisse_seit1949-244692).

3.3 H3 measure for gender equality in the household (in terms of behaviors or outcomes)

Variable that we will use: We will measure gender equality outcomes in a household j as the sum of the absolute differences between the man's weekly hours used for paid work and the woman's weekly hours used for paid work, the man's hours used for housework and the woman's hours used for housework, and the man's hours used for family and the woman's hours used for family:

$$\begin{aligned} TimeDiff_j = \\ |HoursPaidWork_{j,Man} - HoursPaidWork_{j,Woman}| + \\ |HoursHousework_{j,Man} - HoursHousework_{j,Woman}| + \\ |HoursFamily_{j,Man} - HoursFamily_{j,Woman}| \end{aligned} \tag{4}$$

Low values indicate greater support for gender equality, with the maximum support being reached at $TimeDiff_j = 0$. This is the case in households in which the man and the woman share all tasks equally, e.g., both do 40 hours of paid work, do 10 hours of housework and use 20 hours for their family. Note that it is not sufficient for gender equality support that the man and the woman have the same sum of work across these categories. E.g., if the man does 40 hours of paid work and uses no time for housework and family while the woman does no paid work but uses 20 hours for housework and 20 hours for family, $TimeDiff_j = |40 - 0| + |0 - 20| + |0 - 20| = 80$. This is in line with our measure for H1/H4 which measures gender equality attitudes in terms of support for man and woman using the same amounts of time for paid work, housework, and family, respectively. There are some points we would like to note about our measure:

First, within each of the three time use categories, our measure makes no distinction between the man using more time on it than the woman or the other way round. E.g., the measure makes no distinction between households where the man does 20 hours more of paid work than the woman and households where the woman does 20 hours more of paid work than the man. Again, this is in line with our measure for H1/H4 and our understanding of the concept of gender equality.

Second, we can only use our measure for households with a man and a woman, i.e., we cannot use it for same-gender households or single households. This is natural if we want to measure gender equality support in terms of within-household behavior as there is no variation in genders within same-gender households or single households.

Third, we use absolute differences instead of other possible choices such as squared differences to reduce the impact of outliers.

Fourth, given that every person only has a fixed amount of time in their lives, we expect that, in many cases, a paid work difference mechanically translates into housework and family time differences. In such cases our measure includes two sides of the same coin. Yet, even if such a mechanical relationship held for all households, the duplicated differences in our measure would be unproblematic as the measure would merely be a positive linear transformation (multiplying by two) of a measure that, e.g., only includes differences in hours of paid work.

Fifth, the time uses in the SOEP dataset are self-reported which means that the measured time uses are likely different from actual time uses. While random errors would be no problem given a sufficiently large sample, previous studies document systematic errors based on respondents' characteristics. E.g., Yavorsky, Kamp Dush, and Schoppe-Sullivan (2015) compare self-reported time uses of parents with detailed diary entries and

find that men tend to overstate the time they use for housework more than women. While this alone would be no problem for testing H3 (because the error in the time differences between men and women is orthogonal to having daughters), it may be plausible that men and women who become more supportive of gender equality tend to become more similar to each other in terms of time use reporting (e.g., men reduce their overstatement of doing housework). In an extreme example, men who become more supportive of gender equality after having a daughter might increase their time used for housework to the same degree that they reduce their overstatement of doing housework, s.t. we would estimate a zero effect of having daughters on time differences. However, we still believe that our measure can withstand such concerns. Yavorsky, Kamp Dush, and Schoppe-Sullivan (2015) show that errors in self-reported hours of paid work are much smaller which makes sense given that paid work hours are usually contractually defined. Thus, a large fraction of the variation in our measure is likely unaffected by biased self-reporting.

We assign the SOEP time use categories (e.g., https://paneldata.org/soep-core/instruments/soep-core-2015-pe/10) to our three time use categories as follows with the first, second and third variables in parentheses indicating time use on weekdays, Saturdays, and Sundays:

Paid work:

Job/apprenticeship/second job (pl/pli0038_h, pl/pli0003_h, pl/pli0007_h),
Education/further job training (pl/pli0047_h, pl/pli0024_h, pl/pli0028_h).

Housework:

Housework (pl/pli0043_h, pl/pli0012_h, pl/pli0016_h),
Errands (pl/pli0040, pl/pli0054, pl/pli0011),
Repairs/garden work (pl/pli0049_h, pl/pli0031_h, pl/pli0034_v4).

Family:

Child care (pl/pli0044_h, pl/pli0019_h, pl/pli0022_h),
Caring for persons in need of care (pl/pli0046, pl/pli0055, pl/pli0057).

To get a complete picture, we track time use on weekdays and weekends. Only using weekdays would give a distorted picture if parents distribute their tasks, s.t., e.g., the man (woman) does paid work on weekdays (weekends) and takes care of the family on weekends (weekdays). The drawback of including weekend data is that this results in fewer observations as time use on weekdays is recorded every year, while time use on weekends is only recorded every other year (2001, 2003, 2005, 2007, 2009, 2011, 2013, 2015, 2017, 2019) with a total of around 230,000 responses for each time use on weekends. (Note there are also data for 2012 which we will not use because the data only contain

a much smaller fraction of valid responses than the other years, which is why we believe that in 2012, the questions were only asked among a subsample.)

Variables that we considered but will not use: As the literature review in Table 3 shows, direct evidence regarding the effect of daughters on gender support in terms of behavior or outcomes is scarce. Borrell-Porta, Costa-Font, and Philipp (2019) use a dummy that is one if the father does paid work and the mother does not (zero otherwise) as a variable to indicate traditional gender division of work. While we could also use differences in employment status as the dependent variable to test H3 (which we would certainly do as a sensitivity analysis if we could), we prefer differences in time use because of the higher granularity. E.g., in dual-earner households, daughters could still have an effect on gender equality support if the parents of daughters share the housework and family time more equally. Such an effect would be missed in an analysis that only considers the employment status.

3.4 Main independent variables

As outlined in Section 2, our main independent variable is a dummy that is one if a respondent has at least one daughter (zero otherwise), instrumented by a dummy that is one if the firstborn child is a daughter (zero otherwise). To construct these dummies, we use the biobirth/kidsex01, biobirth/kidsex02, etc. variables that indicate the gender of the first, second, etc. child of a respondent. Note that these variables relate to the biological parent-child connection and do not include, e.g., adopted children, which is different from Yang et al. (2025) who also consider non-biological children. This is in line with our interpretation of the hypotheses being about the effect of "having daughters" in the sense of giving birth to a daughter (or in the case of fathers providing half of the genetic material, s.t., the mother gives birth to a daughter). Moreover, as Washington (2008) notes, in the case of adoption, parents can choose the gender of their child which would introduce endogeneity problems into the research design.

3.5 Controls

Because of the exogeneity of the firstborn child's gender, we do not need controls to avoid omitted variable bias. Still, controls are helpful in the estimation of the effects of having daughters as the inclusion of controls can reduce unexplained variation in the dependent variable and reduce standard errors of our coefficients of interest. To make sure that

we capture the full effects of having daughters on the dependent variables, we will only include controls that are determined before the birth of the respondents' first children. For example, the number of children that respondents have may impact their political preferences and the number of children may depend on the first child being a daughter. In this case, having a daughter indirectly affects political preferences via the number of children (i.e., the number of children is a mediator). Including the number of children as a control would exclude this indirect effect from our estimate of the effect of having daughters on political preferences. Because we want to include such an indirect effect in the estimate of the effect of having daughters on political preferences, the number of children (and other variables that could be determined after the birth of the first child such as marital status, religion, education, or work) will not be part of our set of controls.

An additional restriction on controls is that their data availability should not be a limiting factor to our sample size. Therefore, we will only include standard controls (similar to those in Yang et al. 2025) that have few invalid responses: Gender (dummy that is one for males, zero otherwise based on ppathl/sex), age, age squared (with age being determined as the difference between the survey year ppathl/syear and the birth year ppathl/gebjahr), a migration-background dummy (that is one for respondents with a migration background and zero for respondents with no migration background based on ppathl/migback) and survey year fixed effects (for analyses that include more than one survey year, based on ppathl/syear).

3.6 A note on the specific variables that we will use

Sometimes, the SOEP data contain multiple variables for the same concept. E.g., the gender of the first child is recorded via biobirth/kidsex01 and biol/lb0288_h. If we notice that using a variable which we include in this pre-analysis plan results in worse data quality (e.g., more implausible values or fewer observations after merging the different variables) than using another variable that represents the same concept, we will use the variable that results in the better data quality.

4 Inclusion/exclusion criteria

4.1 For all analyses

We will only include observations for parents, which we will determine by an individual having a valid response for biobirth/kidsex01, i.e., the gender of the first child being

either male or female. We will only include observations where the gender of the non-first children (biobirth/kidsex02, biobirth/kidsex03, etc.) is either male, female, or "does not apply" (indicating the individual does not have a second, third, etc. child). We will only include observations with valid responses for the control variables, i.e., ppathl/syear, ppathl/sex, ppathl/gebjahr, and ppathl/migback.

4.2 Additional inclusion/exclusion criteria for tests of H1/H4

We will only include observations with a valid response for pl/plh0308_v2.

4.3 Additional inclusion/exclusion criteria for tests of H2

We will only include observations with a valid response for pl/plh0004.

4.4 Additional inclusion/exclusion criteria for tests of H3

We will only include male-female pairs that live in the same household (identified using ppathl/parid) and that have the same values for having any daughter and for the gender of their first child. We do so because, e.g., if a household is composed of a male with a son and a female with a daughter, it is not clear whether to expect an effect of having daughters for the household as only part of the household has a daughter. On the contrary, if a household is composed of a male and a female that both either have or do not have a daughter and that both either have or do not have a daughter as their first child, it is clearly defined whether to expect an effect of having daughters (even if the children are from different partners). As discussed in Section 3.3, this excludes single parents, same-gender parents and polygamous parents which is necessary to sensibly construct our measure for testing H3.

Moreover, we will only include male-female pairs for which both partners have valid responses for all the time use variables listed in Section 3.3 in a given survey year.

As already noted in Section 3.3, we will only include data from the 2001, 2003, 2005, 2007, 2009, 2011, 2013, 2015, 2017, and 2019 surveys and we will exclude data from the 2012 survey. In 2012, there are much fewer valid responses indicating that the questions were only asked among a subsample.

4.5 Inclusion/exclusion criteria that we will not apply

We interpret the hypotheses as being about the effects of "having daughters" in the sense of being a biological parent to a daughter. This does not require that the daughter lives with the parents. In other words, we will also measure the effects of having daughters for parents whose children have already moved out, for "weekend dads/moms", for parents who have lost custody, and even for parents who have never met their children (e.g., because they gave them up for adoption). This is unlike previous studies (e.g., Yang et al. 2025), that only include respondents who have at least one child living in the same household at the time of the interview. Probably the most common case where our non-application of this filter leads to differences is that we measure the effects of having daughters for parents whose children all have moved out at the time of the interview. We believe that for such parents, there should still be an effect as the experiences of parenthood are lasting and as parents generally still care about their children after they have moved out (in addition to the hypotheses not putting an expiration on the hypothesized effects). Moreover, the exclusion of parents who, e.g., do not have their children live in the same household due to divorce would lead to a selection bias if, e.g., having daughters affects gender equality support which in turn affects divorce rates.

5 Exact analysis and interpretation of results

Inserting the selected variables from Section 3 into the IV regression design from Section 2 yields the regression specifications outlined in the following.

5.1 H1 analysis and interpretation

First stage:

$$\begin{aligned} &AnyDaughter_i = \\ &\pi_0 + \pi_1 FirstbornIsDaughter_i + \pi_2 Male_i + \pi_3 Age_i + \pi_4 Age_i^2 + \pi_5 MigBack_i + \nu_i \end{aligned} \tag{5}$$

Second stage:

$$\begin{aligned} &GenderEqualitySupportAttitude_i = \\ &\beta_0 + \beta_1 \widehat{AnyDaughter}_i + \beta_2 Male_i + \beta_3 Age_i + \beta_4 Age_i^2 + \beta_5 MigBack_i + \varepsilon_i \end{aligned} \tag{6}$$

$GenderEqualitySupportAttitude_i$ indicates on a scale from 1–7 how strongly respondent i agrees that it is best if a man and a woman share the amounts of paid work, housework and family care equally (higher values indicate stronger support for gender equality). $AnyDaughter_i$ is one if respondent i has at least one daughter (zero otherwise). $\widehat{AnyDaughter_i}$ is the predicted value for respondent i from the first stage regression. $FirstbornIsDaughter_i$ is one if the firstborn child of respondent i is a daughter and zero if it is a son. $Male_i$ is one if respondent i is a male (zero otherwise). Age_i is the age of respondent i. $MigBack_i$ is one if respondent i has a migration background (zero otherwise). Standard errors will be clustered by household. Because all data come from the survey year 2018, there will be no survey year fixed effects and standard errors will not be clustered by individual (because there is only one response per individual).

We will confirm H1 ("Having daughters leads to stronger support for gender equality as measured by attitudes.") if and only if our estimate of β_1 is positive and statistically significantly different from zero at the 5% level.

5.2 H2 analysis and interpretation

First stage:

$$\begin{gathered} AnyDaughter_{i,t} = \\ \pi_1 FirstbornIsDaughter_{i,t} + \pi_2 Male_{i,t} + \pi_3 Age_{i,t} + \pi_4 Age_{i,t}^2 + \pi_5 MigBack_{i,t} + \\ \boldsymbol{SurveyYearDummies}_t' \, \boldsymbol{\pi}_{\boldsymbol{SurveyYearDummies}} + \nu_i \end{gathered} \tag{7}$$

Second stage:

$$\begin{gathered} PoliticallyRight_{i,t} = \\ \beta_1 \widehat{AnyDaughter}_{i,t} + \beta_2 Male_{i,t} + \beta_3 Age_{i,t} + \beta_4 Age_{i,t}^2 + \beta_5 MigBack_{i,t} + \\ \boldsymbol{SurveyYearDummies}_t' \, \boldsymbol{\beta}_{\boldsymbol{SurveyYearDummies}} + \varepsilon_i \end{gathered} \tag{8}$$

$PoliticallyRight_{i,t}$ indicates on a scale from 0–10 the left–right self-classification by respondent i in survey year t (higher values indicating being more right-wing).

$AnyDaughter_{i,t}$ is one if respondent i has at least one daughter (zero otherwise) in survey year t. $\widehat{AnyDaughter}_{i,t}$ is the predicted value for respondent i in survey year t from the first stage regression. $FirstbornIsDaughter_{i,t}$ is one if the firstborn child of respondent i is a daughter and zero if it is a son in survey year t. $Male_{i,t}$ is one if respondent i in survey year t is a male (zero otherwise). $Age_{i,t}$ is the age of respondent i in

survey year t. $MigBack_{i,t}$ is one if respondent i in survey year t has a migration background (zero otherwise). $\boldsymbol{SurveyYearDummies}_t$ is a vector of survey year dummies in which, e.g., the dummy for the year 2005 is one if and only if a response is from the survey year $t = 2005$. Standard errors will be clustered by individual and household.

We will confirm H2 ("Having daughters leads to more left-leaning political preferences.") if and only if our estimate of β_1 is negative and statistically significantly different from zero at the 5% level.

5.3 H3 analysis and interpretation

First stage:

$$\begin{gathered} AnyDaughter_{j,t} = \\ \pi_1 FirstbornIsDaughter_{j,t} + \pi_2 Age_{j,m,t} + \pi_3 Age_{j,f,t} + \\ \pi_4 Age^2_{j,m,t} + \pi_5 Age^2_{j,f,t} + \pi_6 MigBack_{j,m,t} + \pi_7 MigBack_{j,f,t} + \\ \boldsymbol{SurveyYearDummies}'_t \, \boldsymbol{\pi}_{\boldsymbol{SurveyYearDummies}} + \nu_j \end{gathered} \tag{9}$$

Second stage:

$$\begin{gathered} TimeDiff_{j,t} = \\ \beta_1 \widehat{AnyDaughter}_{j,t} + \beta_2 Age_{j,m,t} + \beta_3 Age_{j,f,t} + \\ \beta_4 Age^2_{j,m,t} + \beta_5 Age^2_{j,f,t} + \beta_6 MigBack_{j,m,t} + \beta_7 MigBack_{j,f,t} + \\ \boldsymbol{SurveyYearDummies}'_t \, \boldsymbol{\beta}_{\boldsymbol{SurveyYearDummies}} + \varepsilon_j \end{gathered} \tag{10}$$

$TimeDiff_{j,t}$ is the sum of the absolute differences between the weekly hours that the male and the female use for paid work, housework, and family in household j in survey year t. $AnyDaughter_{j,t}$ is one if household j has at least one daughter (zero otherwise) in survey year t. $\widehat{AnyDaughter}_{j,t}$ is the predicted value for household j in survey year t from the first stage regression. $FirstbornIsDaughter_{j,t}$ is one if the firstborn child of household j is a daughter and zero if it is a son in survey year t. $Age_{j,m,t}$ ($Age_{j,f,t}$) is the age of the male (female) in household j in survey year t. $MigBack_{j,m,t}$ ($MigBack_{j,m,t}$) is one if the male (female) in household j in survey year t has a migration background and zero otherwise. $\boldsymbol{SurveyYearDummies}_t$ is a vector of survey year dummies in which, e.g., the dummy for the year 2005 is one if and only if a response is from the survey year $t = 2005$. Standard errors will be clustered by household.

Note that one observation for this analysis corresponds to one household (i.e., one male-female pair) in a given survey year in comparison with all other hypotheses for the tests of which one observation corresponds to one individual in a given survey year.

We will confirm H3 ("Having daughters leads to more gender equality in the household as measured by behaviors or outcomes.") if and only if our estimate of β_1 is negative and statistically significantly different from zero at the 5% level.

5.4 H4 analysis and interpretation

Two first stage equations:

$$\begin{gathered} AnyDaughter_i = \\ \pi_0 + \pi_1 FirstbornIsDaughter_i + \pi_2 Male_i + \pi_3 Age_i + \pi_4 Age_i^2 + \pi_5 MigBack_i + \\ \pi_6 Male_i FirstbornIsDaughter_i + \pi_7 Male_i Age_i + \pi_8 Male_i Age_i^2 + \\ \pi_9 Male_i MigBack_i + \nu_i \end{gathered} \tag{11}$$

$$\begin{gathered} MaleAndAnyDaughter_i = \\ \rho_0 + \rho_1 FirstbornIsDaughter_i + \rho_2 Male_i + \rho_3 Age_i + \rho_4 Age_i^2 + \rho_5 MigBack_i + \\ \rho_6 Male_i FirstbornIsDaughter_i + \rho_7 Male_i Age_i + \rho_8 Male_i Age_i^2 + \\ \rho_9 Male_i MigBack_i + \eta_i \end{gathered} \tag{12}$$

Second stage:

$$\begin{gathered} GenderEqualitySupportAttitude_i = \\ \beta_0 + \beta_1 \widehat{AnyDaughter}_i + \beta_2 Male_i + \beta_3 Age_i + \beta_4 Age_i^2 + \beta_5 MigBack_i + \\ \beta_6 MaleAnd\widehat{AnyDaughter}_i + \beta_7 Male_i Age_i + \beta_8 Male_i Age_i^2 + \\ \beta_9 Male_i MigBack_i + \varepsilon_i \end{gathered} \tag{13}$$

$GenderEqualitySupportAttitude_i$ indicates on a scale from 1–7 how strongly respondent i agrees that it is best if a man and a woman share the amounts of paid work, housework and family care equally (higher values indicate stronger support for gender equality). $AnyDaughter_i$ is one if respondent i has at least one daughter (zero otherwise). $\widehat{AnyDaughter}_i$ is the predicted value for respondent i from the first stage regression for $AnyDaughter_i$. $MaleAndAnyDaughter_i$ is one if respondent i is male and has at least

one daughter (zero otherwise). $\widehat{MaleAndAnyDaughter}_i$ is the predicted value for respondent i from the first stage regression for $MaleAndAnyDaughter_i$. $FirstbornIsDaughter_i$ is one if the firstborn child of respondent i is a daughter and zero if it is a son. $Male_i$ is one if respondent i is a male (zero otherwise). Age_i is the age of respondent i. $MigBack_i$ is one if respondent i has a migration background (zero otherwise). Standard errors will be clustered by household. Because all data come from the survey year 2018, there will be no survey year fixed effects and standard errors will not be clustered by individual (because there is only one response per individual).

We will confirm H4 ("Having daughters leads to a larger increase in the support of gender equality for men than for women with gender equality measured by attitudes.") if and only if our estimate of β_6 is positive and statistically significantly different from zero at the 5% level.

6 Concluding remarks

We are not convinced that pre-analysis plans make research universally more reliable. For example, we usually learn about a complex new dataset by working with it over time (years, not weeks), with our understanding growing through direct engagement. If we had spent years working with the SOEP data, our pre-analysis plan would likely look different.

Our concern is that researchers who deeply understand their data but do not preregister their analyses might have their results dismissed, while those who preregister without fully grasping the data could end up with their work hailed as the standard. Although we agree that pre-analysis plans reduce p-hacking, it is not clear whether they truly benefit the research community, given that understanding complex data is very different from running simple and highly stylized experiments (for a discussion, see, e.g., Olken 2015).

We append a literature overview in table format after the bibliography.

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Table 1: Literature review for H1 and H4

Paper	Details
Yang et al. (2025)	• **Data source:** European Social Survey. • **Dependent variable:** Agreement (1–5): "When jobs are scarce, men should have more right to a job than women.". • **Result:** Small but significant positive effect of daughters on support for gender equality. No difference between men and women.
Wesley and Garand (2021)	• **Data source:** US American National Election Study. • **Dependent variable:** Feminist self-identification, support for women representatives, and support of traditional gender roles. • **Result:** Having sons vs. daughters leads to less support for gender equality, particularly among women.
Borrell-Porta, Costa-Font, and Philipp (2019)	• **Data source:** British Household Panel Survey. • **Dependent variable:** Agreement (1–5): "A husband's job is to earn money; a wife's job is to look after the home and family". • **Result:** Having daughters leads to more support for gender equality among men but not among women.
Perales, Jarallah, and Baxter (2018)	• **Data source:** Household, Income, and Labour Dynamics in Australia Survey. • **Dependent variable:** Composite score from agreement with seven statements regarding gender-role attitudes. • **Result:** Having daughters makes parents more traditional than having sons, fathers in particular.
Sharrow et al. (2018)	• **Data source:** US Cooperative Congressional Election Study. • **Dependent variable:** Composite score from support for three gender-equity policies (Title IX, pay equity, harassment enforcement). • **Result:** Fathers whose firstborn is a daughter (vs. son) are significantly more supportive of gender-equality policies; no systematic effect in terms of the number of daughters.

Paper	Details
Shafer and Malhotra (2011)	• **Data source:** US National Longitudinal Survey of Youth. • **Dependent variable:** Composite score for support of traditional gender roles from eight survey items. • **Result:** Having daughters reduces men's support of traditional gender roles; no significant effect for women.
Warner and Steel (1999)	• **Data source:** Self-collected survey responses from the US. • **Dependent variable:** Support (1–5) for six gender-equity policies (e.g., pay equity, maternity leave, day-care subsidies). • **Result:** Having daughters is associated with higher support for every policy among fathers; effects for mothers are present but weaker.
Warner (1991)	• **Data source:** Self-collected survey responses from the US and Canada. • **Dependent variable:** Support (1–5) for six feminism-related items. • **Result:** Women and Canadian men become more feminist when they have daughters; no significant effect for US men.

Table 2: Literature review for H2

Paper	Details
Yang et al. (2025)	• **Data source:** European Social Survey. • **Dependent variable:** Left-to-right scale. • **Result:** No significant effect of having daughters on political preferences.
Dreber et al. (2024)	• **Data source:** Surveys of members of the Swedish parliament. • **Dependent variable:** Left-to-right scale. • **Result:** No significant effect of having daughters on political preferences.
Clayton, de Kadt, and Dumas (2023)	• **Data source:** South African Social Attitudes Survey. • **Dependent variable:** Binary left-wing vs. right-wing party support. • **Result:** No significant effect of having daughters on political preferences.
Lee and Conley (2016)	• **Data source:** US General Social Survey and European Social Survey. • **Dependent variable:** Binary left-wing vs. right-wing party support & left-to-right scale. • **Result:** No significant effect of having daughters on political preferences.
Conley and Rauscher (2013)	• **Data source:** US General Social Survey. • **Dependent variable:** Binary left-wing vs. right-wing party support & left-to-right scale. • **Result:** Having daughters leads to more right-wing political preferences.
Oswald and Powdthavee (2010)	• **Data source:** British Household Panel Survey. • **Dependent variable:** Binary left-wing vs. right-wing party support. • **Result:** Having daughters leads to more left-wing political preferences.

Paper	Details
Oswald and Powdthavee (2005)	• **Data source:** German Socio–Economic Panel (SOEP). • **Dependent variable:** Binary left-wing vs. right-wing party support. • **Result:** Having daughters leads to more left-wing political preferences.

Table 3: Literature review for H3

Paper	Details
Borrell-Porta, Costa-Font, and Philipp (2019)	• **Data source:** British Household Panel Survey. • **Dependent variable:** Dummy that is one if father does paid work and mother does not (zero otherwise). • **Result:** Daughters lead to less traditional labor division.